

Each card has the same chance. Mix the bags separately. Draw once from each bag, then put both cards back. The bigger number wins.

Problem 1: For each of Problems 1–3, split the pairs among you and circle all nine winners on one shared page. Which deck wins more pairs?

A

2

..

4

:..

9

:.:

B

1

.

6

:..

8

:.:

A

2

..

B

1

.

A

2

..

B

6

:..

A

2

..

B

8

:.:

A

4

:..

B

1

.

A

4

:..

B

6

:..

A

4

:..

B

8

:.:

A

9

:.:

B

1

.

A

9

:.:

B

6

:..

A

9

:.:

B

8

:.:

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Problem 2: Which deck wins more pairs?

B

1

.

6

⋮⋮

8

⋮⋮⋮

C

3

...

5

⋮⋮

7

⋮⋮

B

1

.

C

3

...

B

1

.

C

5

⋮⋮

B

1

.

C

7

⋮⋮⋮

B

6

⋮⋮

C

3

...

B

6

⋮⋮

C

5

⋮⋮

B

6

⋮⋮

C

7

⋮⋮

B

8

⋮⋮⋮

C

3

...

B

8

⋮⋮⋮

C

5

⋮⋮

B

8

⋮⋮⋮

C

7

⋮⋮

Problem 3: Does any of A, B, and C win more pairs than both other decks?

C

3
...

5
...

7
...

A

2
..

4
...

9
...

C

A

C

A

C

A

3
...

2
..

3
...

4
...

3
...

9
...

C

A

C

A

C

A

5
...

2
..

5
...

4
...

5
...

9
...

C

A

C

A

C

A

7
...

2
..

7
...

4
...

7
...

9
...

Problem 4: For each of A, B, and C, choose a deck that wins more card pairs and play six rounds. Must your chosen deck win more rounds?

A	2 ..	4 :..	9 ⋮⋮⋮
B	1 .	6 ⋮⋮⋮	8 ⋮⋮⋮
C	3 ...	5 ⋮⋮⋮	7 ⋮⋮⋮

Problem 5: Try 3, 5, 7, and 9 in A's empty place. Which choices make A win more card pairs than B?

A

2

..

4

⋮

B

1

.

6

⋮

8

⋮

3

...

5

⋮

7

⋮

9

⋮

Problem 6: Swap one A card with one B card. Find every swap that makes B win more card pairs than A.

A	2 ..	4 :..	9 ⋮⋮⋮
B	1 .	6 ⋮⋮⋮	8 ⋮⋮⋮

Problem 7: Put these six cards into two decks of three so both decks win the same number of pairs. Find a way, or show why it cannot be done.

1 •	2 ••	3 •••	4 •••	5 •••	6 •••
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A

--	--	--

B

--	--	--

Problem 8: Can these six cards make three two-card decks where A wins more pairs than B, B more than C, and C more than A? Show a way, or show why it cannot be done.

1 •	2 ••	3 •••	4 •••	5 •••	6 •••
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A

--	--

B

--	--

C

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